

JAYANI SAMARAKOON

Undergraduate in Software Engineering

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PROFESSIONAL SUMMARY

Software Engineering undergraduate with hands-on experience in Machine Learning, AI, full-stack web development, mobile application development, and MLOps. Experienced in building end-to-end AI applications involving data preprocessing, feature engineering, predictive modeling, REST APIs, model deployment, and responsive user interfaces. Familiar with Python, Scikit-learn, PyTorch, TensorFlow, React.js, FastAPI, MLflow, DVC, Docker, and Apache Airflow. Seeking a Software Engineering / AI/ML internship to apply technical skills in real-world projects and contribute to a collaborative development team.

EDUCATIONAL QUALIFICATIONS

BSc (Hons) in Software engineering

2023-Present

Sri Lanka Technology Campus (SLTC)

G.C.E. Advanced Level

Year of 2022

K/MW Pinnwala central college

Technology stream: Engineering technology , Science for technology , Information Technology

SKILLS

TECHNICAL SKILLS

Programming	Python, Java, Kotlin, JavaScript, C, C++, SQL
Frontend	React.js, HTML, CSS
Backend	FastAPI, REST APIs
Mobile	Flutter, Android Studio
AI/Machine Learning	Scikit-learn, Pandas, NumPy ,PyTorch
Databases	MySQL, Firebase
UI/UX	Figma, Wireframing, Prototyping
Tools	Git, GitHub, VS Code, Jira ,Docker,MATLAB,Google Colab,Jupyter

SOFT SKILLS

Team Collaboration & Leadership

Works effectively in teams and takes the initiative to achieve project goals.

Project Management

Plans, organizes, and completes tasks efficiently to meet deadlines.

Learning Agility

Quickly learns new tools, technologies, and techniques to improve performance.

PROJECTS

DeepShield: Explainable Multimodal Framework for Trust Assessment and Evidence-Based Fake News Detection

(Ongoing - Expected Completion: 2027)

- Developing a multimodal AI system using text, image, and metadata for fake news detection.
- Integrating RAG and Explainable AI techniques to provide evidence-based predictions and a Trust Score.
- Building REST APIs using FastAPI and a responsive frontend using React.js.
- **Technologies:** Python, PyTorch, Transformers, BERT, CLIP, OpenCV, FAISS, FastAPI, React.js.

Medical Store Management System

June 2025 – Aug 2025

Responsibilities :

- Designed and developed responsive user interfaces using React.js.
- Created reusable React components for the web application.
- Designed the UI/UX to improve usability and user experience.
- Integrated frontend pages with the backend APIs.
- Collaborated with team members using Agile development practices.
- **Technologies:** React.js/CSS/HTML/JavaScript

End-to-End Customer Churn Prediction with MLOps

Feb 2026 – Mar 2026

Built a production-ready MLOps pipeline that predicts customer churn for Telco customers.

- Cleaned and prepared 7,043 customer records by converting data types, handling missing values, encoding categorical variables, scaling features, and creating an 80:20 train-test split.
- Trained and compared 3 machine learning models (Logistic Regression, Random Forest, and XGBoost), selecting Logistic Regression as the production model with 81.97% accuracy and 0.8621 ROC-AUC.
- **Technologies:** Python, Scikit-learn, MLflow, DVC, Apache Airflow, Docker, FastAPI, DAGsHub

SMS SPAM Detection system

Juny 2026 – July 2026

Responsibilities:

- Developed and trained a Random Forest classifier for SMS spam detection.
- Built and optimized an LSTM deep learning model for sequential text classification.
- Performed feature engineering and hyperparameter tuning to improve model performance.
- Designed and developed a responsive web-based user interface for real-time SMS spam prediction.
- Compared machine learning and deep learning model performance using Accuracy, Precision, Recall, F1Score, and ROC-AUC.
- **Technologies:** Python, Random Forest, LSTM, Flask/FastAPI , HTML, CSS, JavaScript, Scikit-learn, TensorFlow

AI-Based Fruit & Vegetable Detection Web Application

- Developed an AI-powered web application for detecting fruits and vegetables from uploaded images and live camera input.
- Integrated a Roboflow computer vision model to perform object detection and return detected classes, confidence scores, and bounding-box coordinates.
- Built a FastAPI backend with REST API endpoints for image and camera-frame detection.
- Designed a responsive and user-friendly web interface with image upload, live camera detection, loading states, and detection results.
- Implemented HTTPS support and Vercel deployment configuration for web-based access.
- **Technologies:** Python ,FastAPI, Roboflow ,Computer Vision ,HTML,CSS,JavaScript

Hospital Appointment No-Show Prediction Using Machine Learning (Group)

August 2026

Responsibilities:

- Developed a machine learning solution to predict patient appointment no-shows using healthcare data.
- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA).
- Built and evaluated a Logistic Regression classification model.
- Conducted Feature Importance Analysis to improve model interpretability.
- Developed a Streamlit web application for real-time appointment no-show prediction.
- **Technologies:** Python, Pandas, NumPy, Scikit-learn, Matplotlib, Jupyter Notebook, Streamlit, Joblib
- **Live Demo :** <https://smartcare-ai-project-11.streamlit.app/>

Mobile Application Development Project – LankaSmartMart

Jan 2026 – Feb 2026

- Designed the complete UI/UX prototype using Figma, including user-friendly screens and application workflow.
- Developed the Android mobile application using Android Studio by implementing the designed interfaces and core functionalities.
- Integrated the application with the **database** to enable real-time data storage and retrieval.
- Collaborated with team members using Git,GitHub, managing source code through version control and contributing code via commits and pushes.
- Worked closely with the development team throughout the Software Development Life Cycle (SDLC), including planning, UI/UX design, development, testing, and deployment.
- Participated in team discussions, problem-solving, and feature implementation to successfully complete the project.
- **Technologies:** Figma, Android Studio (Java/Kotlin), Firebase/MySQL, Git, GitHub
- **Demo Video:** <https://www.figma.com/proto/yQ4DUNsyD5o3vXHQSZXZ4E/Lanka-Smart-Market?node-id=23&t=e7TEuSFqEJ1UMqQ9-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-nodeid=1%3A2>

Inventory Management & Stock Prediction System

- Developed a full-stack web-based inventory management system using Flask.
- Implemented product, stock, and inventory record management with CRUD operations.
- Designed a responsive and user-friendly dashboard for efficient inventory tracking.
- Integrated SQLite database for reliable data storage and management.
- **Technologies:** Python, Flask, SQLite, HTML, CSS, JavaScript

REFERENCES

Available Upon Request